



MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

MALAWI INSTITUTE OF TECHNOLOGY

APPLIED STUDIES DEPARTMENT

END OF SEMESTER ONE ODEL EXAMINATION

CHEM-1101: GENERAL CHEMISTRY I

DATE: 17th June, 2024.

TIME: 13:30 – 16:30 (3 HRS)

INSTRUCTIONS

1. This examination paper contains **8** questions and **7** pages including this cover page. Please check!
2. Answer **ALL** questions on the answer sheets provided.
3. Write your Reg. Number on each and every page of the answer sheet.
4. Start answering a particular question on a **fresh** page of the answer sheet.
5. All working must be shown clearly.
6. Marks for each part of the question are indicated against it on the right margin.
7. A Periodic Table of the elements and solubility rules have been attached.
8. Phones of any kind are not allowed in the exam room.
9. The following data may be useful:

$$h = 6.63 \times 10^{-34} \text{ Js}, \quad c = 3.0 \times 10^8 \text{ m/s}, \quad R_H = 2.18 \times 10^{-18} \text{ J}$$

(DO NOT TURN OVERLEAF WITHOUT INSTRUCTION)

QUESTION ONE

- a. With relevant examples, differentiate between binary ionic compound and ternary ionic compound. [4]
- b. Name the following compounds:
- i. $\text{Cu}_2(\text{SO}_4)$ ✓ [2]
 - ii. BrF_5 ✓ [2]
 - iii. HClO ✓ [2]
 - iv. $\text{Fe}(\text{NO}_3)_2 \cdot \text{H}_2\text{O}$ [2]

QUESTION TWO

- a. State the Pauli Exclusion Principle ✓ [3]
- b. Write the short hand electron configuration and orbital diagram of Cr^+ . [4] ✓
- c. Write any one possible set of four quantum numbers for an electron in the highest energy orbital for Cr. [2] ✓

QUESTION THREE

- a. The successive ionization energies for an element J are shown in the table below:

Energy number	1st	2nd	3rd	4th	5th
Ionization energy values / kJ mol^{-1}	738	1450	7733	10543	13630

- i. With state symbols, write the equation which represent the first ionization energy of element J. [2]

ii. If element J reacts with chlorine, suggest the formulae of the compound formed. [2]

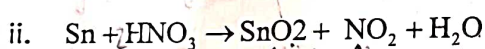
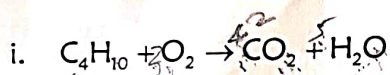
j. Which of the elements, arsenic to rubidium is likely to have the smallest atomic radius and why? [2]

b) Selenium is used in the glass industry to decolorize glass. Draw the orbital diagram for Se. [2]

c) The general trend in electron affinity is that it increases from left to the right across a period. Account for the irregularity in N which has a positive electron affinity or low electron affinity [3]

QUESTION FOUR

a. Balance the following equations:

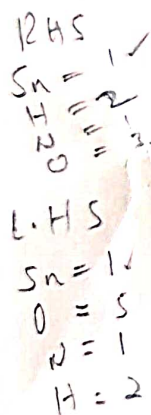
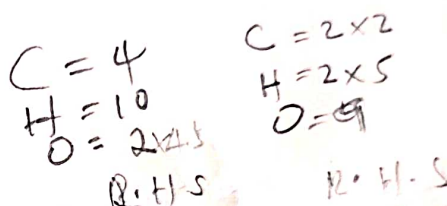
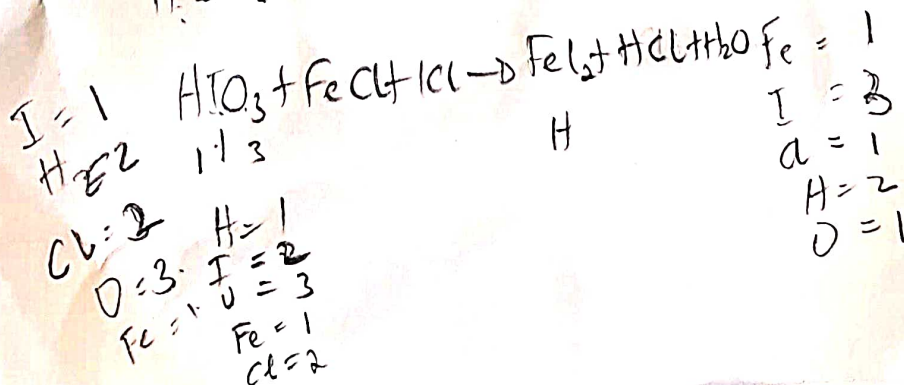
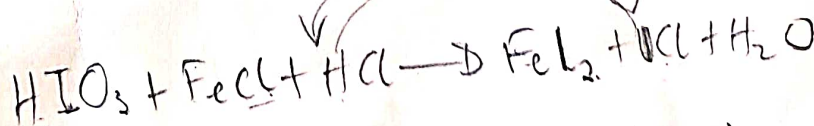
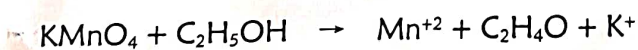


b. Predict the products of the following reactions. In the cases where a precipitate is formed, write a net ionic equation.

i) Lead (II) nitrate reacting with sodium chloride [5]

ii) Calcium chloride reacting with sodium carbonate [5]

c. Balance the following redox reaction in basic medium



QUESTION FIVE

a. Explain the difference between the following:

i. Ionic and covalent bonding. [2]

ii. Covalent bond and dative bond [2]

b. Elements X, Y and Z have 4, 5, and 7 valence electrons respectively. Write the molecular formula of the compounds formed by these elements *individually* with hydrogen. [3]

c. Draw the Lewis structures of the following molecules and state the molecular geometry of (ii) and (iii):

i. CH_3N [2]

ii. O_3 [3]

iii. AlCl_4^- [3]

d. State the types of hybrid orbital of N in the following:

N_2 , NO_2^+ , NO_3^- and NH_4^+ [4]

QUESTION SIX

a. A sample of carbon dioxide CO_2 gas weighs 4.40g. Calculate:

i. The number of moles of CO_2 contained in the 4.40g of CO_2 . [2]

ii. The volume of gas at standard temperature and pressure [3]

iii. The number of molecules of CO_2 contained in 4.40g of CO_2 [3]

iv. The number of C and O atoms contained in the 4.40g of CO_2 [6]

b. How many grams of N_2 (g) can be made from 9.05g of NH_3 reacting with 45.2g of CuO ?



[7]

*** END OF QUESTION PAPER ***

25/24
50-22.4

Copper II sulphide
 Bromine flouride
 Hydroxide

Hydrogen, Chlorine, Oxygen

The Modern Periodic Table of the Elements

Average relative masses are to 2001 values, rounded to two decimal places.
 All average masses are to be treated as measured quantities, and subject to significant figure rules. Do not round them further when performing calculations.

Element name: Mercury
 Symbol: Hg
 Atomic #: 80
 Avg. Mass: 200.59
 Electronegativity: 1.9

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
H 1 1.01	He 2 4.00											B 5 10.81	C 6 12.01	N 7 14.01	O 8 16.00	F 9 19.00	Ne 10 20.18
Li 3 6.94	Be 4 9.01											Al 13 26.98	Si 14 28.09	P 15 30.97	S 16 32.07	Cl 17 35.45	Ar 18 39.95
Na 11 22.99	Mg 12 24.31											K 19 39.10	Ca 20 40.08				
K 19 39.10	Ca 20 40.08											Sc 21 44.96	Ti 22 47.88	V 23 50.94	Cr 24 52.00	Mn 25 54.94	Fe 26 55.85
Rb 37 85.47	Sr 38 87.62											Zn 30 65.39	Ga 31 69.72	Ge 32 72.61	As 33 74.92	Se 34 78.96	Br 35 79.90
Cs 55 132.91	Ba 56 137.33											In 49 114.82	Sn 50 118.71	Sb 51 121.76	Te 52 127.60	I 53 126.90	Xe 54 131.29
Fr 87 (223)	Ra 88 (226)											Tl 81 204.38	Pb 82 207.20	Bi 83 208.98	Po 84 (209)	At 85 (210)	Rn 86 (222)

*lanthanides

La 57 138.91	Ce 58 140.12	Pr 59 140.91	Nd 60 144.24	Pm 61 (145)	Sm 62 150.36	Eu 63 151.97	Gd 64 157.25	Tb 65 158.93	Dy 66 162.50	Ho 67 164.93	Er 68 167.26	Tm 69 168.93	Yb 70 173.04
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**actinides

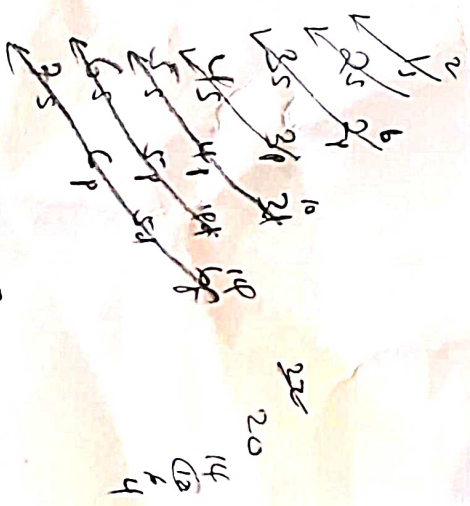
Ac 89 (227)	Th 90 232.04	Pa 91 231.04	U 92 238.03	Np 93 (237)	Pu 94 (244)	Am 95 (243)	Cm 96 (247)	Bk 97 (247)	Cf 98 (251)	Es 99 (252)	Fm 100 (257)	Md 101 (258)	No 102 (259)
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Cr = 6

n = 6

73.8

Sp d s



1s² 2s² 2p⁶ 3s² 3p⁴ 4s² 3d⁵

2p² N + NaCl

Copper

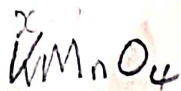
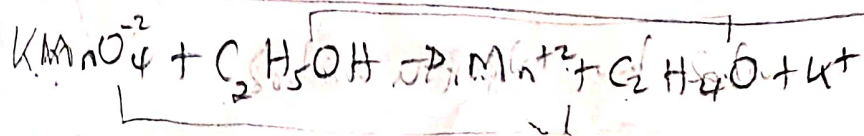
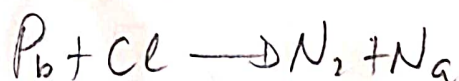
Asenic & Rubidium

Ca + Cl -> NaCO₃

Iron
 Nitrogen
 Oxygen
 Hydroxide

SOLUBILITY RULES

1. Salts of ammonium and Group IA are always soluble.
2. a. All chlorides are soluble except AgCl , Hg_2Cl_2 , and PbCl_2 which are insoluble. b. All bromides are soluble except AgBr , Hg_2Br_2 , HgBr_2 , and PbBr_2 which are insoluble.
c. All iodides are soluble except AgI , Hg_2I_2 , HgI_2 , and PbI_2 which are insoluble.
3. Chlorates, nitrates, and acetates are soluble.
4. Sulfates are soluble except CaSO_4 , SrSO_4 , BaSO_4 , Hg_2SO_4 , HgSO_4 , PbSO_4 , and Ag_2SO_4 which are insoluble.
5. Phosphates, and carbonates are insoluble except NH_4^+ and Group IA compounds.
6. All metallic oxides are insoluble except NH_4^+ and Group IA compounds.
7. All metallic hydroxides are insoluble except NH_4^+ and Group IA and Group IIA from calcium down.
8. All sulfides are insoluble except NH_4^+ and Groups IA and IIA.



$$\begin{aligned} K &= 1 \\ \text{Mn} &= 1 \\ O &= 5 \\ C &= 2 \\ H &= 6 \end{aligned}$$

$$\begin{aligned} \text{Mn} &= 1 \\ C &= 2 \\ H &= 4 \\ O &= 1 \\ K &= 1 \end{aligned}$$

$$\begin{aligned} H &= 8 \\ K &= 1 \\ \text{Mn} &= 1 \\ O &= 6 \end{aligned}$$

$$\begin{aligned} O &= 5 \\ C &= 2 \end{aligned}$$

$$\begin{aligned} O &= 1 \\ C &= 2 \end{aligned}$$

$$\begin{aligned} H &= 6 \\ \text{Mn} &= 1 \\ O &= 2 \end{aligned}$$

